DRELVA VAVE

Six machines, one screen. A Drelva Audio instrument for the M-VAVE FM-1.



Where this stands (10 November 2026). Everything in this manual runs today in the browser emulator, and the screens shown are taken from it. It has not run on an FM-1 yet. Anything that needs the hardware (Bluetooth, the USB audio channels, MIDI clock out, the panel lights, presets kept in the unit's memory) is marked **On the device**: it is designed and has its setting, but it does nothing until the firmware is built for the FM-1. Those sections will be completed after the first hardware test.

Operating manual. Drelva Audio, 2026.

1. What it is

Drelva Vave turns the FM-1 into a six-track techno instrument. Each track holds one **machine**: a sound engine with its own sequencer or pattern, playing by itself in time with the clock. You shape it with the four knobs under the screen, one page of four parameters at a time.

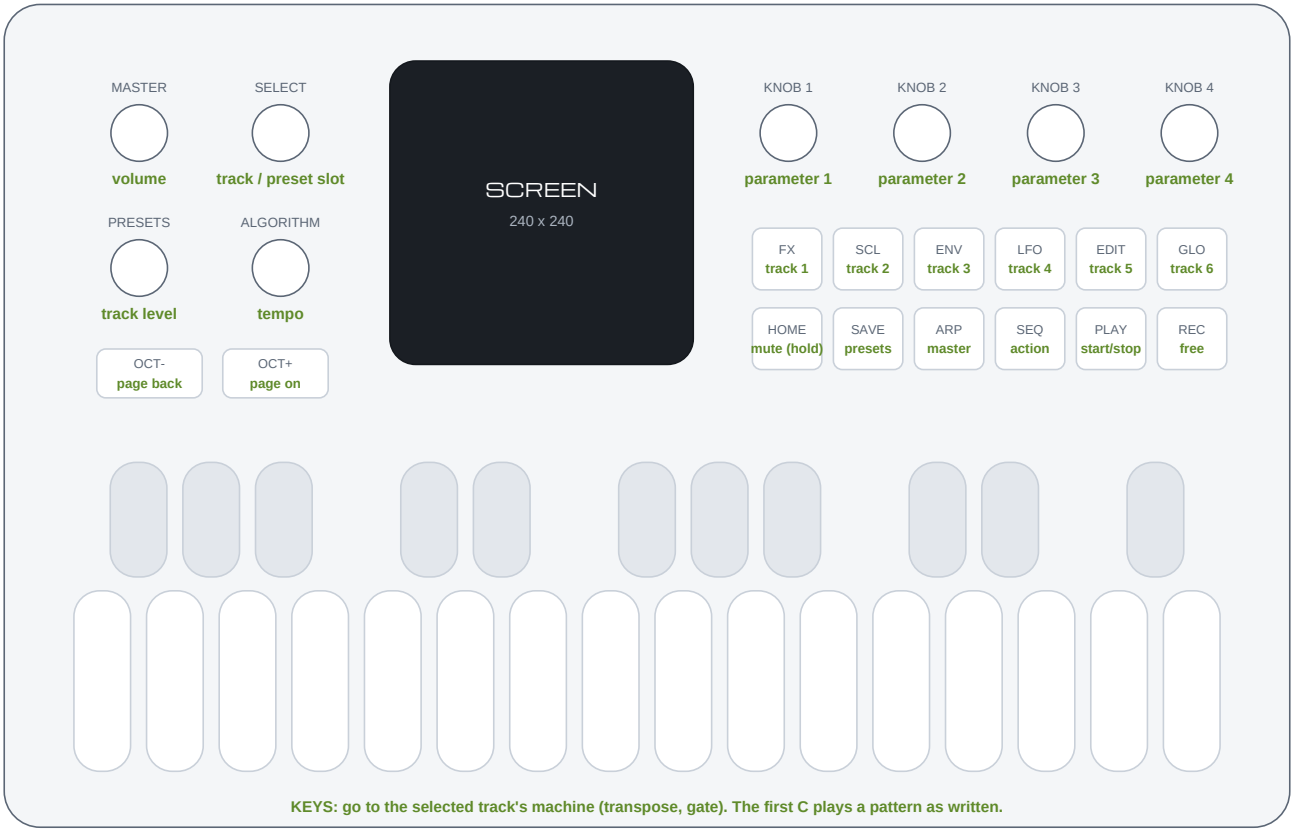
Part	What it does
6 tracks	One machine each. Every track has its own level, mute, and sends to the delay and the reverb.
7 machines	Elastico, Perc Box, Phrase Box, Sismo Box, Stab Box, Swarm Engine, Textura. Any machine can be loaded on any track; each exists once.
Master	Tempo-synced delay, reverb, a one-knob filter and a drive on the stereo mix.
Presets	8 slots. A preset holds everything: which machine is on each track, every parameter, the mix, the master and the tempo.
Clock	Its own tempo (60 to 200 BPM), or it follows incoming MIDI clock.
USB audio	**On the device.** Each track on its own channel, dry, plus the stereo mix with the effects.

Tracks when you switch on: 1 Sismo Box, 2 Elastico, 3 Textura, 4 Swarm Engine, 5 Phrase Box, 6 Perc Box. Stab Box is not on a track at first; load it where you want it (section 5).

The switch-on settings are tuned to work together as techno at 134 BPM: everything sits in G (the kick at G1, the elastic line, the drone and the stabs on G), the kick is the loudest thing, the Master ducks the other tracks on every beat, and Phrase Box marks every 8 bars with a different transition.

2. The panel

The names printed on the FM-1 belong to its original firmware. In grey: the printed name. In green: what the control does in Drelva Vave.



Control	Does	While PRESETS is open
PLAY	Start and stop. Starting always begins every pattern from its first step.	the same

Control	Does	While PRESETS is open
FX SCL ENV LFO EDIT GLO	Select track 1 to 6. With HOME held: mute or unmute that track.	-
ARP	Master section. Again: System. Again: back to the track you were on.	-
SELECT	Step through tracks 1 to 6, Master, System.	Pick the slot, 1 to 8
OCT- / OCT+	Previous / next page of what is selected.	OCT- closes; OCT+ loads the slot
KNOB 1 to 4	The four parameters shown on the page. Turn fast to move in bigger steps.	the same
ALGORITHM	Tempo, 60 to 200 BPM. Does nothing while an outside clock is followed.	the same
PRESETS	Level of the selected track.	the same
SEQ	The machine's action: a new pattern, or a hit (see each machine).	-
SAVE	Opens Presets.	Stores into the slot and closes
HOME	Held with a track button: mute.	-
REC	Not used.	-
MASTER	Output volume.	the same
Keys	Sent to the selected track's machine.	the same

3. The screen



Area	Shows
Top line	Name of the selected machine. Right: tempo. **BPM** means its own clock; **EXT** in green means it is following an outside clock and the number is the measured tempo. After a preset is stored or loaded the name is replaced for a moment by SAVED, LOADED or EMPTY.
Four bars	The beat. The lit bar is the current quarter of the bar.
1 to 6	The tracks. Filled: selected. The small bar is its level meter. **M**: muted. All six outlined in green: Master or System is selected.
Picture	The machine's own display: its pattern and where it is playing.
Page line	Page name, one mark per page (the lit one is where you are), and the track's level.
Four dials	The parameters of KNOB 1 to 4: value in the middle, name below. A dial that goes below zero grows from the top. A dark dial has no parameter.

4. Playing

To	Do this
Start	Press PLAY. All machines start together on step 1.
Work on a track	Press its button (FX = 1 ... GLO = 6) or turn SELECT.
Change the sound	Turn KNOB 1 to 4. OCT+ and OCT- move through the machine's pages.
Mute	Hold HOME and press the track's button. Again to unmute.
Set a level	Turn PRESETS for the selected track, or use LEVEL on its MIX page.
Change tempo	Turn ALGORITHM.

To	Do this
Get a new pattern	Press SEQ on Elastico or Sismo Box. On Phrase Box and Stab Box, SEQ fires a hit.
Change key	Select a melodic track and press a key. The first C (eighth key from the left) is the pattern as written; other keys move it up or down by semitones. (On the two STAGE pages of Elastico the white keys pick a stage instead.)

Each machine keeps what it is doing when you look at another track. Knobs only ever act on what the screen shows.

5. The MIX page and loading machines



The last page of every track is **MIX**. It is the same on all six.

Knob	Does
LEVEL	Track level (the same value PRESETS turns).
DELAY	How much of the track goes to the master delay.
REVERB	How much goes to the master reverb.
LOAD	Which machine the track holds. Turn it to load another one, or NONE to empty the track.

Each machine exists once. If you load a machine that another track is holding, the two tracks **change places**: the other track gets the machine you had. Nothing is lost, and every machine keeps its settings while it is not on a track.

Order on the dial: NONE, ELAST, PERC, PHRASE, SISMO, STAB, SWARM, TEXTUR.

6. The machines

In alphabetical order. Each table lists the pages in the order OCT+ reaches them; MIX always comes after.

Elastico



The elastic voice of Parkoid without its click, for hypnotic lines. An FM pair into a filter, bent by a spring: every note plucks the spring, and its swing stretches the pitch and, with FLEX, the brightness, then snaps back and settles. The spring starts from rest and is pushed, so the pitch leaves the note smoothly instead of jumping, and the level fades in over SOFT: that is why it does not click. It is played by an eight-stage sequencer in the manner of the ML-185.

Gate modes. REST: silent for all its pulses. SINGLE: one note on the first pulse. MULTI: a note on every pulse. HOLD: one note held through all the pulses.

MODE. FWD 1 to LAST. REV backwards. PEND there and back. RANDOM any stage. DRUNK mostly forward, sometimes staying or stepping back.

Keys. On the STAGE pages the first eight white keys pick the stage to edit. On every other page the keys transpose. **SEQ** rolls a whole new pattern using the GEN page; it replaces all eight stages, which you can then edit. Presets keep the stages. The PARKER scale is mostly the root, with the odd octave, fifth, flat seventh and flat third: a figure rather than a tune.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
ELASTIC	STRETCH how far the pitch swings, in semitones; negative swings down first	SPEED how fast the spring swings	DAMP how soon it comes to rest	FLEX how much the swing also opens the tone
TONE	RATIO of the FM pair: 0.5 to 7	INDEX FM amount	FOLD wavefolder	CUTOFF
SHAPE	SOFT attack, from sharp to 60 ms	DECAY	RESO	GLIDE between notes
MOVE	DRIFT slow change of FM amount and filter	RATE of the drift: 4, 8, 16 or 32 bars	TUNE semitones	-
STAGE	STAGE which stage you are editing, 1 to 8	PITCH semitones, -12 to +24	PULSES how long the stage lasts, 1 to 8	GATE Rest, Single, Multi, Hold
STAGE 2	STAGE	SLIDE the note glides in without a new attack	ACCENT a harder pluck	SKIP the stage is passed over
SEQ	LAST how many stages play, 1 to 8	MODE Fwd, Rev, Pend, Random, Drunk	DIV a pulse is 1/16, 1/8 or 1/4	GATE note length, % of a pulse
GEN	SCALE Parker, minor, Phrygian, pentatonic, octaves	OCTS 1 or 2	SLIDE % chance per stage	ACCENT% chance per stage

Reading the picture. Eight columns, one per stage. Left of each: the pitch slider (a white cap is an accent). Right: eight lamps, the first pulse at the bottom; lit lamps are the stage's pulses, bright where a note starts, joined for HOLD, hollow for REST. A line under a stage is a SLIDE. Greyed stages are skipped or beyond LAST. The frame marks the stage being edited; the white lamp is the pulse playing now, and the cap of the playing stage rides the spring.

Starting points. Rubber: STRETCH 9, SPEED 45, DAMP 40. Slow wobble under the beat: SPEED 15, DAMP 20. A tight zip: STRETCH 18, SPEED 80, DAMP 75. Falling instead of rising: negative STRETCH. For a figure that never repeats with the bar, set LAST to 5 or 7.

Perc Box



Hat, clap and a metallic FM perc in one track. Each has its own Euclidean pattern: HITS are spread as evenly as 16 steps allow, and ROTATE turns the pattern round the bar.

Picture: three rows of 16 steps, H, C and P. White: playing now.

Useful starts: hat 4 hits rotated 2 is the off-beat; 16 hits is straight sixteenths. Clap 2 hits rotated 4 falls on beats 2 and 4. SEQ and the keys do nothing here.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
HAT	HITS 0 to 16	ROTATE 0 to 15	DECAY	OPEN how long the off-beat hats ring
CLAP	HITS	ROTATE	DECAY tail	TONE
PERC	HITS	ROTATE	PITCH semitones	DECAY
SHAPE	HAT TON brightness	FM amount	RATIO 1.41, 2.00, 2.76, 3.50	ACCENT how much the strong steps stand out
LEVELS	HAT	CLAP	PERC	-

Phrase Box



Counts phrases and plays a transition at every phrase line: a build before it, a hit on it, a fall after it. Leave it running and the track marks its own sections.

FX: HIT, RISE, R+HIT, ROLL (snare roll), DOWN (downlifter), SWELL (reverse cymbal), SIREN, KICKS (kick roll), ZAP, SUB (sub drop), ALL, or VARY, which picks a new combination every phrase.

Picture: the next transition by name, the step count, and the phrase as a bar with a mark where the build starts. **SEQ:** a hit now.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
PHRASE	PHRASE length in steps: 16, 32, 64, 128, 256	FX which transition	LEAD how many steps the build and the fall take: 4 to 64	RNDLEN with VARY: also vary the lead
RISE	RESO of the sweep	TONE noise to saws	PITCH semitones	TAIL of the hit
HIT	CRASH noise in the hit	DRIVE	-	-

Sismo Box



Kick and rumble, after the Sismo plugin. A synthesized kick on every beat, and under it a low burst on every sixteenth, shaped by a groove, filtered, saturated and ducked by the kick.

Picture: the groove, 16 bars grouped by beat; a dot under each kick. White: playing now.

SEQ: switches GROOVE to GEN and rolls the next one. The keys do nothing here.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
RUMBLE	RUMBLE level	GROOVE Roll, Offbeat, Gallop, Triplet, Warehouse, or GEN (generated)	WEIGHT how dark or open	GRIT saturation
KICK	PUMP how hard the kick ducks the rumble	PITCH semitones	DECAY of the kick	TAIL of the rumble
PUNCH	CLICK attack	SNAP depth of the pitch drop	SEED which generated groove	-

Stab Box



Dub techno chord stabs: four saws on a chord, a resonant filter that pings open on each stab. Not on a track at switch-on; load it with LOAD. It wants the delay: turn up DELAY on its MIX page.

Picture: the chord as four bars, the four bars of the progression, and the rhythm.

SEQ: a stab now. **Keys:** move the root.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
CHORD	CHORD Min, Min7, Min9, Dub, Sus4, 5th, Min11	RHYTHM Off, Dub, Push, 3/16, One, Man (only SEQ)	PROG chord per bar: Stay, i-VI, i-iv, i-VII, Fall	ROOT semitones
FILTER	CUTOFF	ENV how far the stab opens it	DECAY	RESO
VOICE	DETUNE	-	-	-

Swarm Engine



A drone of 16 oscillators: four voices at an interval, each a centre oscillator and three detuned ones.

MODE: PLAY sounds while the transport runs. KEYS sounds only while a key is held. SYNC is PLAY with every oscillator restarted on each beat.

Picture: one ring per voice with its four oscillators going round. **Keys:** set the pitch.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
SWARM	TUNE semitones	SPREAD detune	INTVL unison, octaves, fifths, fourths, cluster	WAVE saw, pulse, fold, XOR
SHAPE	FOLD (for WAVE fold)	MODE Play, Keys, Sync	TONE low-pass on the whole swarm	MOVE slow sweep of TONE, one turn every four bars

Textura



The air around the groove: a band of noise whose colour drifts, a low hum and some crackle, all pushed down on every beat and rising back, so it breathes with the kick.

It sounds while the transport runs, and while a key is held. **Picture:** the band, low to high, sinking on each beat.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
TEXTURE	COLOR where the band sits, low to high	WIDTH wide and soft to narrow and whistling	DRIFT how far the colour wanders	RATE how fast
MOTION	PUMP how far each beat pushes it down	RELEASE how slowly it comes back	CRACKLE	HUM

7. Master



Press **ARP**. The Master works like a track: three pages, four knobs. Only the stereo mix goes through it; the separate track outputs stay dry.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
DELAY	TIME 1/16, 1/8, 3/16, 1/4, locked to the tempo	FEEDBK repeats	TONE how dark the repeats get	RETURN delay level
REVERB	SIZE	DAMP	RETURN reverb level	-
OUTPUT	FILTER left of centre closes a low-pass; right opens a high-pass; centre is off	DRIVE	PUMP how far each beat ducks every track except the one holding Sismo Box	-

What reaches the delay and the reverb is set per track, on its MIX page. With all sends at 0 the effects are silent whatever RETURN says. PUMP is the sidechain of a techno mix; it acts on the stereo mix only, so the separate track outputs are not ducked. The picture shows the six track meters and, on the right, the left and right output.

8. Presets



To	Do this
Open	Press SAVE. The eight slots appear; a dot marks the ones in use.
Pick a slot	Turn SELECT.
Store	Press SAVE again. The top line says SAVED. A slot in use is overwritten without asking.
Load	Press OCT+. The top line says LOADED, or EMPTY if the slot has nothing.
Leave	Press OCT-.

A preset holds: the machine on each track, every parameter of those machines, level, sends and mute of each track, the Master, and the tempo. It does not hold the System settings.

Loading while playing is fine; patterns carry on from where they are.

In the emulator, presets are kept by the browser on that computer. **On the device.** They will be kept in the FM-1's memory; that part is not written yet.

9. System settings

Press **ARP twice** (once for Master, again for System), or turn SELECT all the way right. These settings belong to the instrument, not to a preset. They are written to memory when you leave System.



CLOCK



LINK



PANEL

Page	Setting	Values	What it does
CLOCK	SOURCE	AUTO, INT	AUTO follows MIDI clock whenever it arrives and goes back to its own tempo one second after it stops. INT ignores outside clock. Works now.
	OFFSET	-40 to +40 ms	Moves the instrument against the clock it follows. Negative plays earlier: use it to take out the delay of a Bluetooth link. Works now.
	SEND	OFF, ON	**On the device.** Sends its own MIDI clock, Start and Stop.
LINK	BT MIDI	OFF, ON	**On the device.** Switches Bluetooth MIDI on. This is where Bluetooth is turned on; see section 10.
	USB OUT	STEREO, TRACKS	**On the device.** What the computer sees: the stereo mix only, or six tracks plus the mix (8 channels). See section 11.
	BITS	16, 24	**On the device.** Bit depth of the USB audio.
PANEL	BRIGHT	10 to 100	Screen brightness. The emulator dims its screen with it.
	KEY LED	OFF, ON	**On the device.** Lights in the keys.

The picture on this screen is a status board: whether the clock is its own or followed, what the USB output is set to, and whether Bluetooth MIDI is on.

10. Clock and Bluetooth

With SOURCE on AUTO, Drelva Vave follows MIDI clock from any input. The tempo display turns green and reads EXT. MIDI Start makes it play from step 1, Stop stops it, Continue carries on. Its own PLAY still works. ALGORITHM is ignored while it follows.

A Bluetooth clock does not arrive evenly. Drelva Vave does not play each tick as it comes: it measures the tempo over the last two beats, runs its own steady clock at that tempo, and corrects itself gently towards the master. Measured in the emulator with a simulated uneven clock:

Master tempo	Ticks late by up to	Kick wanders by	Average lateness
134 BPM	0 ms	0.15 ms	1.3 ms
126 BPM	10 ms	0.9 ms	6 ms
140 BPM	20 ms	1.6 ms	11 ms
90 BPM	30 ms	2.2 ms	15 ms

The average lateness is about half the unevenness, and it is steady, so OFFSET removes it: for a link that is up to 20 ms late, try OFFSET -10. With a very uneven clock the first two or three beats after Start can be off by a few tens of milliseconds before it settles.

On the device. Bluetooth is not active yet. The plan: set LINK > BT MIDI to ON; the FM-1 then appears as a Bluetooth MIDI device; connect to it from the sending device's Bluetooth MIDI list (on a Mac: Audio MIDI Setup; on an iPhone or iPad: the Bluetooth MIDI screen of the app). The base this firmware is built on has been used that way with a Mac and an iPhone by its author; a Windows PC has not been tried. The exact pairing steps, and what the real unevenness of the link is, will be written here after the first test on hardware. Until then, clock can be tried in the emulator through the MIDI in menu under the drawn panel.

11. USB audio

On the device. Not active yet. This is the layout the firmware is designed to present, with USB OUT on TRACKS.

Channel	Carries
1 to 6	Tracks 1 to 6, mono, after the track's level and mute, without delay, reverb, filter or drive.
7 and 8	The stereo mix, left and right, with the Master effects: what the headphone output plays.

The channels follow the tracks, not the machines: if you load another machine on track 3, channel 3 carries it. In a DAW, make six mono audio tracks with inputs 1 to 6, and one stereo track on 7/8 if you want the effects. Do not monitor 7/8 together with 1 to 6, or the dry sound is heard twice.

On Windows a device with more than two channels usually needs an ASIO layer for Ableton Live to see them all (ASIO4ALL or FlexASIO). How many channels Windows accepts from the FM-1 without a driver is not known yet; if it is fewer than eight, the fallback is the stereo mix over USB and recording tracks one at a time.

12. Quick reference

Printed on the FM-1	Drelva Vave	Printed on the FM-1	Drelva Vave
PLAY	Start / stop	KNOB 1 to 4	Page parameters
FX	Track 1	SELECT	Track / preset slot
SCL	Track 2	ALGORITHM	Tempo
ENV	Track 3	PRESETS	Track level
LFO	Track 4	MASTER	Volume
EDIT	Track 5	OCT-	Page back / close presets
GLO	Track 6	OCT+	Page on / load preset
HOME + track	Mute	SAVE	Presets / store
ARP	Master, then System	SEQ	New pattern / hit
REC	Not used	Keys	Transpose / gate

13. Credits and licence

Drelva Vave is free software under the GNU General Public License, version 3. Whoever gives you the firmware must also offer you its source code.

Its hardware layer is built on Melodee by Kerem Kilic (Ellic Studio), itself based on Felucca by Leo Kuroshita and Hugelton Instruments. The page of the browser emulator is adapted from Melodee's, with the web emulator build by David Konsumer. The machines come from Drelva Audio's Drelvarom firmware, except three new ones: Elastico (after the elastic voice of the Parkoid plugin), Perc Box and Textura. The typeface is Michroma (SIL Open Font License).

M-VAVE and FM-1 are trademarks of their owner. Drelva Vave is not made, endorsed or supported by M-VAVE. Installing other firmware on an FM-1 is at your own risk: the unit has a single firmware area, and an update that is interrupted can leave it unable to start.